

## ABSTRACT

Statistical arbitrage on equity pairs rests on 1 assumption: that 2 historically correlated stocks will revert to a common spread after a temporary divergence. This paper documents an empirical study of pair selection on the Nikkei 225 universe, comparing 2 approaches under identical signals, position sizing, and transaction costs: classical industry classification, and unsupervised k-Means clustering on return correlations. Industry classification produces a portfolio total return of +1.24% (avg pair Sharpe 0.083, 3/5 pairs winning); k-Means produces −0.42% (avg pair Sharpe 0.002, 1/5 pairs winning).

The study surfaces a failure mode that cuts across both strategies. The 1 overlapping pair — Sojitz (2768.T) and Sumitomo (8053.T), both *sogo shosha* — scored highest on cointegration strength (ADF: −3.90,  $p < 0.01$ ) and correlation (0.969), yet produced the worst back-test result once a copper-mine acquisition by Sumitomo structurally decoupled the spread. This failure is selection-method-agnostic: it cannot be diagnosed from price data because it originates in a corporate news event.

The 2nd contribution is a proposed experimental design for an LLM news-screening layer whose sole job is to flag pair-breakdown events before they destroy capital. The claim is narrow and testable: does interposing an LLM news filter between the z-score signal and position sizing reduce drawdown during structurally-driven spread divergences, relative to a pure-quant baseline?

## 1. Introduction

A pair trade produces a signal — an entry z-score — and the system must decide whether to act on it. In most implementations the answer is mechanical: if  $|z| > \text{threshold}$ , enter; if  $|z| < \text{exit-threshold}$ , close. This works when spread divergences are mean-reverting noise. It fails when they are structural breaks in disguise.

A structural break looks, in the 1st days, exactly like a normal widening. Price data does not separate the cases. What separates them — often — is a news event: an acquisition, a regulatory change, a commodity price shock specific to 1 leg. This information exists in text, not in the cointegration statistic — and is not solved by choosing a better pair-selection method. Both industry classification and clustering will select a pair if the statistical signal is strong enough.

LLMs can read financial news and judge whether a specific corporate event is likely to affect the long-run relationship between 2 companies. The question this paper proposes to test: is that judgment, applied at entry time, profitable enough to justify the added complexity?

## 2. Data and Universe

|          |                                                                |
|----------|----------------------------------------------------------------|
| Universe | 188 N225 constituents, avg. daily volume > 1 M shares          |
| Period   | January 2023 – May 2025 · 589 trading days                     |
| Source   | Yahoo Finance (yfinance)                                       |
| Lookback | 90-day rolling window · min. 60 common days (holiday-adjusted) |

## 3. Pair Selection

### Strategy 1 — Industry Classification

Candidate pairs are formed within industry groups. Of 63 N225 industry groups, 23 contain only 1 constituent and are excluded. All within-industry combinations of the remaining groups are scored and ranked. Rationale: shared macro drivers, supply-chain exposure, and regulatory environment provide a fundamental basis for cointegration. Limitation: cross-sector pairs with strong statistical cointegration are structurally excluded.

## Strategy 2 — k-Means Clustering

|                   |                                                    |
|-------------------|----------------------------------------------------|
| <b>Input</b>      | 90-day rolling daily return matrix (stocks × days) |
| <b>Algorithm</b>  | k-Means++ with elbow method · optimal k = 8 (N225) |
| <b>Pairs from</b> | Intra-cluster combinations only                    |

Clustering groups stocks by geometric similarity of return trajectories, regardless of industry. Of the top-5 pairs selected by each strategy, 1 pair overlapped: Sojitz (2768.T) vs. Sumitomo (8053.T).

### Scoring Formula and Trading Parameters

Score = ADF weight + 2.0 × |Pearson r| [ ADF weight: 3 (1%) · 2 (5%) · 1 (10%) · 0 (none) ]. Top-5 pairs selected.

|                                |                                                                          |
|--------------------------------|--------------------------------------------------------------------------|
| <b>Spread · Z-score window</b> | Price1 - Price2 · 30-day rolling                                         |
| <b>Entry / Exit thresholds</b> | z  > 1.5 entry ·  z  < 0.4 exit · 1-day signal lag                       |
| <b>Capital · Costs</b>         | JPY 1,000,000 per pair · 0.15% per leg (entry + exit) · 50/50 long-short |

## 4. Results

| Metric                 | Strategy 1 — Industry | Strategy 2 — k-Means |
|------------------------|-----------------------|----------------------|
| Portfolio total return | +1.24%                | −0.42%               |
| Avg pair Sharpe        | 0.083                 | 0.002                |
| Pairs winning          | 3 of 5                | 1 of 5               |
| Pair overlap           | —                     | 1 of 5               |

*Avg pair Sharpe is the mean of the 5 selected pairs' individual Sharpe ratios — not the Sharpe of the combined portfolio equity curve, which is a distinct quantity. Because each pair is measured in isolation, this metric ignores cross-pair diversification, so the combined-portfolio Sharpe is computed separately and generally differs.*

Industry classification outperforms k-Means by 1.66 pp on the same signal, same capital, and same transaction-cost assumption. Within-industry pairs share a denser set of common fundamental drivers — regulation, supply chains, customer bases — that support tighter, more durable spreads. The cross-sector pairs discovered by k-Means co-moved statistically in the formation window but proved less robust in the back-test, with only 1 of 5 pairs profitable. The result does not settle the general question — the back-test is short and the sample small — but it establishes that statistical co-movement alone is not a sufficient basis for durable cointegration.

## 5. The Sojitz / Sumitomo Failure

|                        |                                                               |
|------------------------|---------------------------------------------------------------|
| <b>Pair</b>            | Sojitz (2768.T) / Sumitomo (8053.T) — both <i>sogo shosha</i> |
| <b>ADF / Pearson r</b> | −3.90 (p < 0.01) · r = 0.969                                  |
| <b>Score</b>           | Highest of all pairs under both strategies                    |
| <b>Outcome</b>         | Worst back-test result in the portfolio under both strategies |

Both companies are diversified Japanese trading houses with overlapping commodity, energy, and industrial businesses. The statistical relationship was extraordinarily strong in the formation window. During the back-test period, Sumitomo announced a material copper-mine acquisition — shifting its commodity exposure, changing its capital allocation profile, and permanently altering the long-run relationship between the 2 companies' earnings streams. The spread did not revert.

### Key insight

The information that this divergence was structural was in news text at the time of signal generation — not in prices. This failure is selection-method-agnostic: industry classification did not protect against it; k-Means did not protect against it. Neither can, because neither reads news. This is the gap the proposed LLM layer addresses.

## 6. Proposed Architecture: LLM News Screening

The LLM layer intercepts the signal pipeline at 1 decision point: between z-score threshold breach and position entry. It answers 1 question: *Does recent news about either leg of this pair suggest the spread divergence may be structural rather than mean-reverting?*

| Step | Component       | Description                                |
|------|-----------------|--------------------------------------------|
| 1    | Z-Score Signal  | $ z  > 1.5$ threshold breached             |
| 2    | LLM News Filter | Query prior 7 days of news for each ticker |
| 3    | PASS / FLAG     | Structural break detected?                 |
| 4    | Position Entry  | Enter if PASS · suppress if FLAG           |

LLM inputs: ticker identifiers, news headlines + summaries (prior 7 days), 1-line description of the assumed fundamental relationship. Output: PASS or FLAG, plus an optional 1-sentence reasoning trace.

### What Gets Flagged

| FLAG — Potential Structural Break         | PASS — Expected to Mean-Revert                |
|-------------------------------------------|-----------------------------------------------|
| M&A / acquisition; divestiture / spin-off | Earnings beat/miss; analyst upgrade/downgrade |
| Regulatory action specific to 1 leg       | Dividend announcement                         |
| Commodity exposure shift; credit event    | Macro events affecting both legs equally      |

Implementation: financial newswire API (headlines + summaries sufficient); instruction-following LLM with financial domain competence; 1 query per entry signal (O(hundreds of tokens) — rounding error vs. trading costs).

## 7. The Proposed Experiment

Null hypothesis: adding an LLM news-screening layer produces no statistically significant improvement in risk-adjusted return over the pure-quant baseline.

|                                 |                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Strategy 1+ (Baseline)</b>   | Industry classification + z-score signal · no news filter · stronger performer on the back-test period |
| <b>Strategy 1-L (Treatment)</b> | Identical pair selection and signal · LLM screening at entry · flagged signals suppressed              |
| <b>Validation</b>               | Walk-forward on a held-out period not used in pair-formation or scoring calibration                    |

| Metric                    | Why It Matters                                       |
|---------------------------|------------------------------------------------------|
| Net return (1-L vs. 1+)   | Does screening add alpha?                            |
| Max drawdown (1-L vs. 1+) | Does screening reduce tail events?                   |
| Win rate on flagged pairs | Were flagged pairs actually structural breaks?       |
| False-positive rate       | How often did screening suppress profitable entries? |
| Sharpe ratio (1-L vs. 1+) | Risk-adjusted comparison                             |

### Honest baseline

If Strategy 1-L does not outperform Strategy 1+ on the held-out period net of costs, the LLM layer should be removed. A reasonable success target: the layer prevents 2–3 Sojitz/Sumitomo-style events per year per 5-pair portfolio, without suppressing more than 10–15% of profitable entries.

## 8. Discussion

Most proposed LLM applications in finance fall into 1 of 2 failure modes:

- **The forecasting mistake** — using LLMs to predict price direction, with no principled advantage over quantitative methods.
- **The integration mistake** — proposing LLMs as a general orchestration layer with no clear, testable claim.

This proposal avoids both. The task — classifying whether a news event is a structural break — gives LLMs a genuine comparative advantage. Price data cannot read a news article; an LLM can. The classification boundary is well-defined enough to evaluate against ground truth.

The LLM layer is pair-selection-agnostic. Whether pairs are chosen by industry grouping or k-Means, any pair that triggers an entry signal is a candidate for screening. The narrow claim: the LLM helps at the 1 moment when the signal fires, by reading news that the quantitative model cannot read.

## 9. Limitations

|                               |                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Back-test is small</b>     | 1 589-day period on 1 equity universe. A 1.66 pp differential is directional, not deployment-grade.         |
| <b>LLM layer not tested</b>   | Section 6 is a proposed design. Claims about LLM effectiveness are conditional on empirical validation.     |
| <b>News coverage uneven</b>   | Large-cap N225 names have adequate coverage. Smaller names or non-Japanese markets may suffer sparsity.     |
| <b>Language / translation</b> | Japanese-language news may require a multilingual LLM or translation step — additional cost and error risk. |
| <b>LLM hallucination</b>      | Model may flag non-events or miss real events. False-positive and false-negative rates must be measured.    |

## 10. Conclusion

We adopted k-Means clustering to move past simple industry classification toward a more sophisticated, data-driven method of pair discovery — 1 that groups stocks by the geometry of their return trajectories rather than by sector labels. On the N225 universe over the study period, that bet did not pay off: industry classification outperformed k-Means by 1.66 pp, consistent with the hypothesis that shared fundamental drivers, not statistical co-movement alone, underpin durable spread relationships. The more sophisticated selection method did not beat the simpler, fundamentals-based one.

Neither strategy protects against the failure mode identified here: a structurally-driven spread divergence caused by an idiosyncratic corporate news event. The Sojitz/Sumitomo pair — the highest-scoring pair under both strategies — failed for exactly this reason. The information needed to avoid the loss was in the news, not in the prices.

The proposed LLM news-screening layer addresses this gap at 1 decision point, with a narrow and testable claim. If the LLM layer does not improve risk-adjusted return on the held-out period, it should not be deployed.

## References

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